

Rotational constraints on nuclear population memory in a cryptochrome reaction model

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Abstract

A molecular history can affect a later response only if it survives the motions and reactions between preparation and readout. Spin-selective chemistry can prepare nuclear populations, but their retention in cryptochrome depends on the molecular environment. Here we show, within a two-nucleus reaction model, that free rotation removes a history contrast that survives at fixed orientation. With identical interaction tensors and reaction rates, the contrast after 20 preparation cycles is unresolved at a reporting resolution of 10^{-10} for all three tested free-rotation times. Protecting only the hydroquinone state is insufficient. Reversible binding restores a conditional contrast: at an assumed bound fraction of 0.50000 and mean dwell time of 1.00000 s, increasing the root-mean-square wobble angle from 1.00000 to 5.00000 degrees reduces the contrast from 1.28861×10^{-3} to 1.36135×10^{-5} . Removing coherence at cycle boundaries reproduces these slow-cycle yields, consistent with population inheritance studied previously. Electric orientation does not rescue the freely rotating history; fixed-geometry electronic response and deformation of an assumed binding well modify the bound-state contrast. A synthetic stopping circuit illustrates possible downstream amplification under prescribed gain, without identifying a native neural pathway. These results replace a single storage-time interpretation with conditional constraints on motion, chemical renewal and readout. Testing a biological memory requires measurements of binding, angular fluctuations and local transduction in the same preparation.

Keywords: cryptochrome, nuclear spin populations, rotational diffusion, quadrupolar relaxation, molecular binding, electric field

1 Introduction

A biochemical system can respond to its past only when a physical variable carries information from one event to another. Unequal spin-state populations are one possible variable: a spin-selective reaction prepares a distribution, subsequent dynamics modify it, and a later reaction reads it through a changed yield. Whether such a history can reach a cell depends on more than the lifetime of an isolated spin state. The molecule moves, changes chemical state and interacts with its environment before the next readout. The relevant question is which of those operations preserve an observable difference.

Cryptochrome motivates this question because electron–nuclear spin dynamics can influence radical-pair reaction yields [1, 2], and magnetic sensitivity has been measured in purified European-robin cryptochrome 4 [3]. Experimental precedents for preparing nuclear populations include a flavin–tryptophan model compound [4] and engineered light–oxygen–voltage proteins [5]; a wider review describes the diversity of flavoprotein spin dynamics [6]. These systems establish that spin-selective chemistry can prepare nuclear polarisation. They do not establish a common storage lifetime across chemical states, proteins or cellular compartments.

Population inheritance across cryptochrome reactions also has a direct theoretical precedent. Wong and colleagues propagated joint nuclear populations between photocycles after removing coherences in the nuclear Zeeman basis [7]. Their equation (8) retains joint diagonal probabilities, including diagonal correlations. Their largest enhancements of directional sensitivity required favourable retention and radical configurations; more realistic cases showed smaller enhancements. Thus, population inheritance itself is not a new principle, and a favourable calculation does not establish biological performance. Their directional-anisotropy observable differs from the fixed-condition history contrast studied here.

Motion is central to assessing that contrast. Fluctuating interactions cause relaxation in radical-pair models, as demonstrated by molecular-dynamics-based calculations for cryptochrome electron spins [8]. Closed-shell chemical states remove unpaired-electron hyperfine interactions but retain nuclear interactions. In particular, spin-1 ^{14}N couples through its electric quadrupole moment to the electric field gradient (EFG) [9]. Rotation of an anisotropic interaction can therefore affect a nuclear distribution even during a closed-shell hydroquinone (HQ) interval. A fixed-orientation calculation cannot determine how much history survives free rotation, temporary binding or fluctuations within a binding site.

An electric field introduces several further distinctions. Torque on an effective molecular dipole changes the distribution of molecular orientations. Work on model transmembrane peptides provides a precedent for voltage-dependent molecular reorientation [10], but does not identify cryptochrome’s local field or effective dipole. Electronic redistribution can separately change the EFG and hyperfine coupling (HFC), and the same dipole torque can deform an assumed mechanical restraint. Preferential orientation does not imply complete immobilisation, and neither is equivalent to spin polarisation. These mechanisms must be separated before an electric-field-dependent yield can be interpreted.

The possibility of neural readout supplies a motivation, rather than a calibration, for this analysis. In *Drosophila*, cryptochrome-dependent light responses affect neuronal firing [11], involve the redox-sensitive potassium-channel subunit Hyperkinetic [12], and can differ from circadian resetting [13]. Light-driven reactive-oxygen-species production by *Drosophila* cryptochrome [14], cofactor-dependent current modulation in a mammalian Kv channel preparation [15], and a lipid-peroxidation-associated channel memory in fly neurons [16] concern different steps and preparations. They motivate chemical and channel readouts, while leaving a dark nuclear-history-to-neuron conversion unmeasured. Local production, transport and removal constrain redox signals [17]; release stochasticity and synapse-to-synapse variation further constrain neural transmission [18].

However, a fixed-orientation storage result does not determine whether a history remains after a complete reaction sequence with finite rotation, reversible binding and electric perturbations. In this study, we keep the reference interaction tensors, magnetic field and reaction conditions fixed while varying these physical conditions. We find that free rotation removes the prepared contrast, whereas sufficiently persistent binding with sufficiently small fluctuations retains a conditional contrast. We then separate electric orientation, electronic susceptibility and binding-well deformation, and use previously generated synthetic circuit trials to illustrate the consequences of selected molecular inputs. The result is a set of model-specific retention conditions and falsifiable measurement requirements, not a prediction that cryptochrome controls stopping in the brain.

2 Methods

2.1 Reaction model and provenance

The model retains two spin-1 flavin nuclei, N5 and N10, with a nine-dimensional nuclear Hilbert space. H denotes anionic closed-shell HQ, R the closed-shell oxidised state, E a neutral semiquinone escape state, and RP the radical pair. E and RP have Hilbert-space dimensions 18 and 36. A triplet-mixture electron preparation is combined with the nuclear state on entering RP. The magnetic field is $50.00000 \mu\text{T}$; the singlet-product and escape rates are 10^7 and $4 \times 10^6 \text{ s}^{-1}$. Product recovery, reduction and HQ waiting have rate 10 s^{-1} , and escape oxidation has rate 4 s^{-1} . These rates define scenarios and were not fitted to neural observations. Additional oxygen-electron relaxation, electron exchange and electron dipolar coupling are set to zero, which is a restrictive assumption.

The product branch returns through R; the escape branch returns through E and electron tracing. A further R reduction interval and an H interval complete the cycle. The complete map and rate assignments are given in Online Resource 1 (Sections S1–S2). The nuclear state initially is $I_9/9$. After 20 completed preparation cycles, the next product yield is evaluated. This protocol does not fix elapsed preparation time and is not an assertion of a stationary state or a passive storage lifetime.

The reference Hamiltonians and local dissipators are frozen from the existing electronic-structure and trajectory calculation. The local bath uses a 6 ps oxidised-state trajectory at 298 K with a 1 ps covariance window; transferring that bath

across states and extrapolating its Markovian generator are assumptions. The same frozen input is used in each motion comparison. Spin-selective reaction loss follows the Haberkorn construction [19]. More complete reaction theories can add reactive exchange and dephasing [20]. Local and reduced wobble dissipation use positive-coefficient Lindblad forms [21]. Online Resource 1 specifies which inputs are calculated, imposed or unmeasured.

2.2 Full, population and reset interventions

For each physical condition, three propagation rules are compared. The full model retains the density matrix. The population model applies

$$\mathcal{D}_H(\rho) = \sum_j P_j \rho P_j, \quad P_j = |j\rangle\langle j|, \quad (1)$$

at the H cycle boundary in the local H Hamiltonian eigenbasis; in the bound sector, this is the well-averaged Hamiltonian. This operation retains joint populations and diagonal nuclear correlations. It removes inter-cycle coherence, while preserving coherent spin dynamics within each cycle. Population refers to $|c_j|^2$ for a pure state, not the complex amplitude c_j . Consequently this comparison is neither an amplitude-only wavefunction model nor a removal of all quantum dynamics.

The reset control replaces the nuclear state by $I_9/9$ at the same boundary while preserving its orientation and binding marginal. We define the history contrast

$$\Delta q = q_{\text{full}} - q_{\text{reset}}. \quad (2)$$

Both yields use the same current field, rates and motion assumptions. Reset still includes the single-cycle spin-selective reaction. In binding calculations, repeated reset can indirectly change the inherited motion distribution. We therefore additionally reset only the final full-state nuclear distribution, decomposing Δq into nuclear and motion-marginal contributions (Online Resource 1, Section S3). Full-population agreement tests an observable under a specified boundary operation; it is not equality of the entire states or proof of a universally classical mechanism.

2.3 Free rotation and reversible binding

Common rigid-body rotational diffusion is propagated throughout the cycle rather than redrawing orientation at each chemical transition. In the body frame, with $\mathbf{n} = R^{-1}\hat{\mathbf{z}}$, the rotational part of the stochastic Liouville generator is

$$\mathcal{G}_{\text{rot}} = -D_R \sum_{a=x,y,z} (L_{\text{orb},a} + \text{ad } J_a)^2, \quad D_R = (6\tau_2)^{-1}, \quad (3)$$

where $\text{ad } J(\rho) = [J, \rho]$ and J includes the spins present in the chemical state. We test $\tau_2 = 9.72000, 21.47660$ and 99.12000 ns. These are selected conditions, not an inferred brain distribution. An exponential chemical wait of rate r acts through the resolvent

$r(rI - \mathcal{G})^{-1}$. Separate interventions suppress rotation in H alone or in all H/R/E waiting intervals, or accelerate all chemical waits while leaving RP reaction rates fixed.

Reversible binding adds spin-conserving capture and release in every chemical state, including RP. The free rotational time is 21.47660 ns. Prescribed motion-only bound fractions are 0.01000, 0.10000, 0.50000 and 0.90000; dissociation rates are 0.01000, 1.00000 and 100.00000 s^{-1} . The resulting mean dwell times are the inverses of these rates. The bound fraction during a reaction cycle need not equal the prescribed motion-only equilibrium fraction. Angle-selective capture controls are also evaluated.

Bound fluctuations are represented by a small-angle, fast-noise reduction. The root-mean-square (RMS) rotation-vector angle is 0.00000, 0.10000, 1.00000 or 5.00000 degrees; it is not a cone half-angle. For component variance $v = \theta_{\text{rms}}^2/3$ and correlation time τ_w , the additional dissipator is

$$\mathcal{G}_w = 2v\tau_w \sum_a \mathcal{D}[V_a], \quad V_a = -i[J_a, H_{\text{body}}], \quad \mathcal{D}[V]\rho = V\rho V^\dagger - \frac{1}{2}\{V^\dagger V, \rho\}. \quad (4)$$

Hamiltonians are in angular-frequency units. Six positive-weight rotations approximate the mean Hamiltonian and local bath. The reference $\tau_w = 10.00000$ ps is supplemented by selected 100.00000 ps cases. This is an effective generator, not an explicit simulation of Ornstein–Uhlenbeck trajectories. Treating local and added rotational noise as independent could double count unresolved fluctuations.

2.4 Three electric-field mechanisms

First, molecular orientation is biased by an assumed dipole potential $U_E/(k_B T) = -\kappa \cos \theta$. The illustrative maximum field 14.00000 MV/m, effective dipole 1000 D and temperature 310 K give $\kappa = 10.91094$, with electric and magnetic fields parallel. Neither the dipole nor the local field is measured for the proposed cryptochrome population. The bath remains the frozen 298 K input; these are controlled field comparisons, not a fully temperature-matched native ensemble.

Second, static linear EFG and HFC susceptibilities are calculated with the same geometries, electronic states and MM point-charge environments as the zero-field tensors. H/R use a 12 Å embedding radius and def2-SVPD; E uses the full minimum-image environment and uncontracted pcJ-2 on nitrogen. The calculations use PySCF 2.13.0 [22], B3LYP, grid level 4 and density fitting. Analytic coupled-Kohn–Sham response is checked against central finite-field SCF differences in the direction $(1, 2, 3)/\sqrt{14}$. The third-rank susceptibility is transported through the saved local rotations before constructing $H(E)$. Both contact and spin-dipolar HFC are included. A uniform external field has no direct gradient; the calculated EFG change comes from electronic redistribution. Geometry relaxation, MM polarisation, field-dependent reaction rates and changes in the intrinsic local noise spectrum are excluded.

Third, a conditional mechanical well is deformed using

$$\frac{U(R)}{k_B T} = \frac{c}{2} |\log R|^2 - \kappa \mathbf{n} \cdot R\mathbf{p}, \quad c = 3/\theta_{\text{rms},0}^2. \quad (5)$$

Here R is the local rotation relative to an anchor, $\mathbf{p}$ the assumed unit dipole axis and $\mathbf{n}$ the field direction in that frame. The minimum and tangent-space Hessian K give covariance K^{-1} . Fixed rotational mobility yields correlation times $\tau_i = c\tau_0/\lambda_i$, where λ_i are Hessian eigenvalues. This restraint is not a calculated native binding free-energy surface. With uniform anchor orientations and equal free/bound dipoles, the exact orientational partition function factorises, leaving the total motion-only equilibrium bound fraction unchanged. The small Laplace normalisation error is removed; it is not interpreted as a binding effect. Dissociation remains 1.00000 s^{-1} . Four matched arms at $\kappa = 10.91094$ isolate orientation alone, added electronic response, added well deformation and both additions.

2.5 Conditional neural readout and numerical verification

Selected zero-electric-field binding yields are used in a synthetic circuit with 64 stochastic connections and eight selected input connections. A finite pool of 10000 product units is prepared once, then relaxes with an assumed 200.00000 ms lifetime. A saturating conversion with gain 1024 modifies release probability on a stop route; temporal filtering and delayed trigger/stop processes determine whether stopping precedes motor onset. This is an illustrative gain choice, not a CRY-to-synapse measurement. Online Resource 1 (Section S6) gives the probability law, architecture, time discretisation and classifiers. Race models motivate separating initiation and cancellation [23, 24]; empirical stop-signal guidance stresses detection and triggering limitations [25].

Each of 18 molecular conditions has 24576 paired trials, generated as three seeds with 8192 trials each. Four classifiers read the same trials; they are not independent biological replicates. The main article uses the temporal classifier. Conservative intervals for the paired probability difference combine exact binomial intervals for positive and negative discordances with Bonferroni coverage of at least 95% under the simulated trial model. They quantify Monte Carlo uncertainty conditional on all imposed parameters. The electric-field-updated yields have not been propagated to these circuit simulations.

Angular cutoff, quadrature, independent resolvent and spectral checks are reported separately from physical-model uncertainty. Values unresolved below 10^{-10} are reported as such rather than as meaningful machine-scale differences. This threshold is a reporting resolution, not a certified error bound. Five decimal places in numerical displays are a formatting rule; stored inputs retain full precision. Online Resource 2 regenerates figures and tables and checks arithmetic and intervals from frozen results. It does not rerun all upstream molecular dynamics, electronic structure or neural trials.

3 Results

3.1 Free rotation removes the slow-cycle history contrast

The fixed-orientation reference has $\Delta q = 0.00570$ and a calculated isolated-storage half-decay time of 1.63474 s. These are conditional controls. With finite free rotation, full, population and reset yields instead agree within the 10^{-10} reporting resolution at

all three tested rotational times (Table 1). The inverse slowest nontrivial H decay rate ranges from 2.33398 to 22.28120 μs (Fig. 1a). These generator eigenvalue diagnostics are not measured T_1 values, observable-specific half-decay times or behavioural times.

Table 1 Free rotation leaves no resolved slow-cycle history contrast. Cycle yields use angular cutoff $L = 3$; spectra use the highest archived cutoff for each condition. Absolute-yield convergence is assessed separately in Online Resource 1.

τ_2 (ns)	q_{full}	q_{reset}	Inverse rate (μs)	$ \Delta q $
9.72000	0.35068	0.35068	22.28120	$< 10^{-10}$
21.47660	0.34151	0.34151	10.20655	$< 10^{-10}$
99.12000	0.30933	0.30933	2.33398	$< 10^{-10}$

Protecting H alone does not restore a resolved contrast. Protecting all H/R/E waiting states while allowing RP to rotate gives $\Delta q = 1.71094 \times 10^{-4}$ (Fig. 1b). Thus, the loss cannot be assigned only to the final H interval: other chemical waits erase the distribution that would reach it. The all-waits-protected preparation has a separate added-storage half-decay time of 1.53143 s; its state and observable differ from the fixed-orientation reference.

Accelerating all waits provides a different counterfactual. Multipliers 10^4 , 10^6 and 10^8 produce contrasts 2.88347×10^{-9} , 1.41208×10^{-5} and 2.00984×10^{-4} , respectively. At 10^8 , the mean preparation time is only 3.37123 μs and the full-population difference is 2.80920×10^{-5} . The slow-cycle population approximation therefore need not hold for rapid renewal. These extreme rates do not identify a physiological operating regime. Nor do they provide long passive retention: adding free H storage suppresses the prepared contrast (Online Resource 1, Section S2).

The changed absolute yields under rotation should not be attributed wholly to H relaxation, because RP dynamics and the orientation ensemble also change. Likewise, a single reset-prepared probe retains a 0 versus 50.00000 μT yield difference of 7.92616×10^{-4} in the tested protocol. Magnetic response in one reaction and retention of a previous reaction’s history are different observables.

3.2 Binding duration and angular fluctuations jointly control retention

Reversible binding yields a graded family of retained contrasts (Fig. 2). At motion-only bound fraction 0.50000 and mean dwell time 1.00000 s, Δq decreases from 2.24704×10^{-3} without wobble to 1.28861×10^{-3} at 1.00000° and 1.36135×10^{-5} at 5.00000° (Table 2). At 1.00000° , reducing the dwell time to 0.01000 s gives 2.12761×10^{-6} . Binding occupancy alone therefore does not specify retention; exchange and angular noise must also be stated.

At a 1.00000 s dwell time and 1.00000° wobble, raising the prescribed bound fraction from 0.01000 to 0.90000 increases Δq from 2.51286×10^{-5} to 2.35779×10^{-3} .

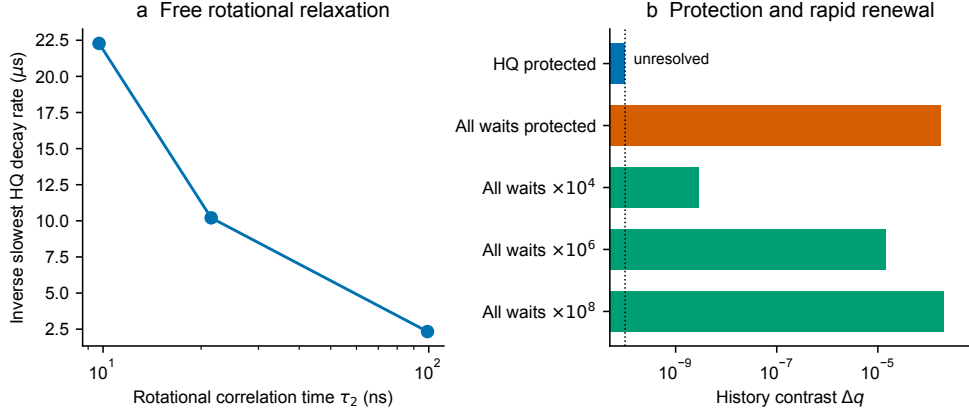


Fig. 1 History retention requires protection across the cycle or sufficiently rapid renewal in this model. (a) Free-rotation H spectra. The inverse decay rate is a generator diagnostic. (b) Matched protection and wait-rate interventions. The dotted line is the 10^{-10} reporting resolution; the H-only bar terminates at that line to indicate an unresolved contrast. Rate multipliers apply to chemical waits, not to RP reaction rates.

Increasing the wobble correlation time from 10.00000 to 100.00000 ps at fraction 0.50000 reduces the contrast from 1.28861×10^{-3} to 9.13516×10^{-5} . No universal monotonicity with wobble amplitude follows: the 0.10000° example slightly increases the 1.00000 s result relative to zero wobble. Mean-Hamiltonian changes and relaxation both contribute.

Table 2 History contrasts for reversible binding at motion-only fraction 0.50000 and $\tau_w = 10.00000$ ps. Each entry uses the highest available angular cutoff: $L = 3$ for 0.00000, 1.00000 and 5.00000°; $L = 2$ for 0.10000°.

RMS angle (°)	Dwell 100.00000 s	Dwell 1.00000 s	Dwell 0.01000 s
0.00000	3.46702×10^{-3}	2.24704×10^{-3}	1.91602×10^{-6}
0.10000	3.43841×10^{-3}	2.25135×10^{-3}	1.92191×10^{-6}
1.00000	1.66289×10^{-3}	1.28861×10^{-3}	2.12761×10^{-6}
5.00000	1.46687×10^{-5}	1.36135×10^{-5}	6.57789×10^{-7}

Across the archived binding sweep, the maximum absolute full-population yield gap is 4.90795×10^{-11} . The retained readout in these slow-cycle cases is therefore consistent with joint population inheritance. It does not require coherence across the selected H boundary. For the 1.00000°, 1.00000 s, fraction-0.50000 case, final-state nuclear reset resolves a nuclear contribution of 1.29817×10^{-3} and a motion-marginal contribution of -9.55628×10^{-6} . Most of the contrast in this example is nuclear, while the decomposition prevents a motion-distribution change from being mislabelled as pure spin memory. No passive seconds-scale lifetime has been calculated for the binding sweep.

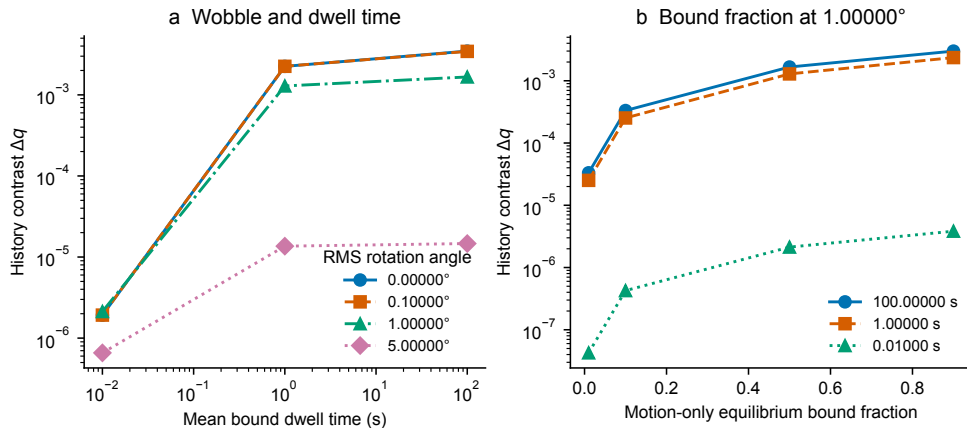


Fig. 2 Reversible binding retains history conditionally on dwell time and bound fluctuations. (a) Four RMS rotation-vector angles at motion-only bound fraction 0.50000. (b) Three dwell times at 1.00000°. All curves use $\tau_w = 10.00000$ ps and isotropic capture. Lines connect computed scenarios, not a fitted biological law. The highest available angular cutoff is used for each point; source tables retain the cutoff.

3.3 Electric orientation does not substitute for immobilisation

At the strongest assumed orienting potential, $\langle \cos \theta \rangle = 0.90835$ and $\langle P_2(\cos \theta) \rangle = 0.75025$. Despite this strong preference, the free-population history remains unresolved. The H inverse slowest rate decreases from 10.20655 to 6.16071 μ s. Even a limiting one-axis alignment retains axial rotational relaxation, with inverse rates of 1.65261–4.86319 μ s for the tested alignments. Orientation is therefore insufficient for protection in these cases.

For bound molecules at 1.00000°, fraction 0.50000 and dwell time 1.00000 s, orientation alone changes Δq from 1.28861×10^{-3} to 1.12030×10^{-3} , a reduction of 13.06128% in the matched $L = 2$ comparison. This calculation assumes an effective dipole and field rather than inferring either from a membrane voltage measurement.

3.4 Electronic susceptibility and well deformation produce distinct corrections

At 14.00000 MV/m, the maximum directional linear EFG changes are 0.02199–0.04879% of the corresponding frozen QM EFG norm (Fig. 3a). For neutral SQ, the maximum isotropic HFC changes are 0.01143 MHz at N5 and 0.00887 MHz at N10, compared with zero-field values 11.03879 and 6.12097 MHz. EFG and HFC maxima need not occur in the same direction. The finite-field derivative checks disagree with analytic derivatives by less than 0.15000% in the tested direction; this is a numerical cross-check within the selected electronic model, not a measure of its accuracy in a protein ensemble.

The mechanical response depends on anchor direction. At initial RMS angle 5.00000°, a parallel well narrows to 4.95488°, an antiparallel well broadens to 5.04725°,

and a perpendicular well shifts its mean orientation by 1.58632° . Uniformly distributed anchors and identical free/bound dipoles leave the total equilibrium fraction unchanged in this model. A native increase in binding or a slower dissociation rate cannot be inferred without a binding partner and transition-state model.

Table 3 Factorial electric comparison at the same orienting potential $\kappa = 10.91094$, bound fraction 0.50000, dwell 1.00000 s and reference $\tau_w = 10.00000$ ps. All entries use $L = 2$ to keep the four-arm comparison matched. Relative changes are calculated before rounding.

Initial RMS ($^\circ$)	Added mechanism	Δq	Change (%)
1.00000	Orientation only	1.12030×10^{-3}	0.00000
1.00000	Electronic	1.12005×10^{-3}	-0.02242
1.00000	Well	1.12128×10^{-3}	0.08728
1.00000	Both	1.12103×10^{-3}	0.06485
5.00000	Orientation only	1.24033×10^{-5}	0.00000
5.00000	Electronic	1.23949×10^{-5}	-0.06744
5.00000	Well	1.35026×10^{-5}	8.86289
5.00000	Both	1.34935×10^{-5}	8.78920

With orientation already imposed, electronic response alone changes the bound contrast by -0.02242% at 1.00000° and -0.06744% at 5.00000° (Table 3). Well deformation gives $+0.08728\%$ and $+8.86289\%$; the combined changes are $+0.06485\%$ and $+8.78920\%$ (Fig. 3b). The larger relative correction occurs where the baseline contrast is small: the combined 5.00000° contrast remains 1.34935×10^{-5} . The combined $L = 3$ result is 1.34934×10^{-5} ; this last displayed-digit change is recorded rather than mixed into the $L = 2$ factorial table.

The maximum full-population difference across these runs is 3.68989×10^{-11} . For the freely rotating population, adding electronic response still leaves no resolved history. At fixed positive orienting potential, positive and negative electronic-response fields give H inverse rates 6.14517 and 6.17628 μs , respectively, compared with 6.16071 μs without electronic response. This sign reversal is a factor-isolation control, not reversal of the entire membrane potential.

3.5 Neural effects remain conditional on the imposed conversion

In the selected temporal circuit readout, the no-wobble, fraction-0.50000, dwell-1.00000 s input increases stopping by 5.50944 percentage points, with interval [5.17006, 5.84444]. At 1.00000° this becomes 3.38135 points, [3.11008, 3.64891], and at 5.00000° it becomes 0.05697 points, [0.01057, 0.10164] (Fig. 4). A 0.01000 s dwell time at 1.00000° gives 0.00814 points, $[-0.01715, 0.03305]$. Equal-input free-rotation trials give zero paired change. Their nonzero conservative interval reflects interval construction, not observed discordance.

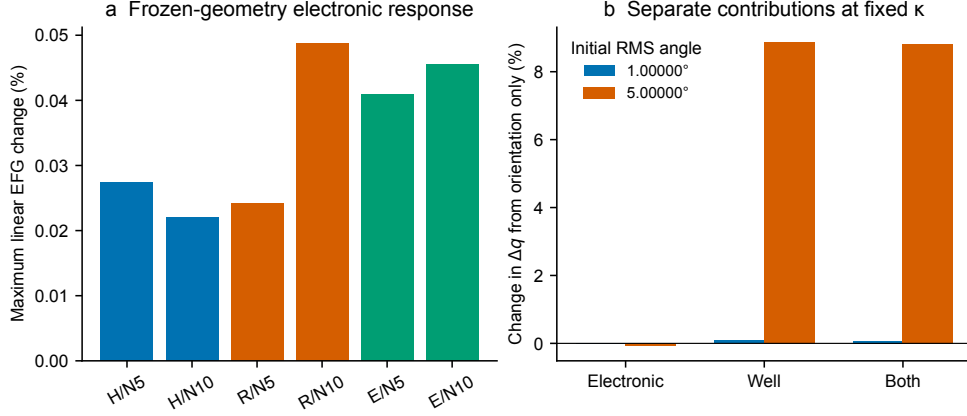


Fig. 3 Electronic response and deformation of an assumed binding well affect different parts of the model. (a) Maximum directional EFG response at 14.00000 MV/m, normalised to each frozen QM EFG Frobenius norm. H/R are closed shell; E is neutral SQ. (b) Relative yield-contrast changes at fixed $\kappa = 10.91094$ and $L = 2$. “Well” denotes the conditional rotational restraint, not an identified native binding energy.

These are outputs of the assumed gain, source pool, selected connections and timing. They do not measure synaptic release changes caused by cryptochrome. For a single preparation of N independent binary product units, any common downstream stochastic readout satisfies

$$|\Delta P| \leq 1 - (1 - |\Delta q|)^N, \quad (6)$$

by coupling the two preparations and applying the same subsequent dynamics [26]. If $|\Delta q|$ were bounded by 10^{-10} , $N = 10000$ would imply at most 0.00010 percentage points. Here the premise uses a numerical reporting scale, not a certified physical upper bound. Repeated independent writing events, different downstream kernels or correlated product sources require a different bound. No updated neural effect is claimed for the electric-field arms.

4 Discussion

The central result is the dependence of a reaction-readable nuclear history on motion throughout the chemical cycle. An isolated fixed-orientation lifetime is insufficient to characterise a freely rotating molecule. Protecting only H leaves other waits available to erase the distribution, whereas protecting all waits or maintaining a sufficiently persistent bound population permits a nonzero contrast. The binding calculations supply a conditional continuum between those limits. They do not identify which point describes a particular neuron.

Population dominance has a precise scope here. The slow-cycle yields are reproduced after removal of H-boundary coherence, including when finite rotation, binding and electric corrections are present. A stochastic population map at that boundary can

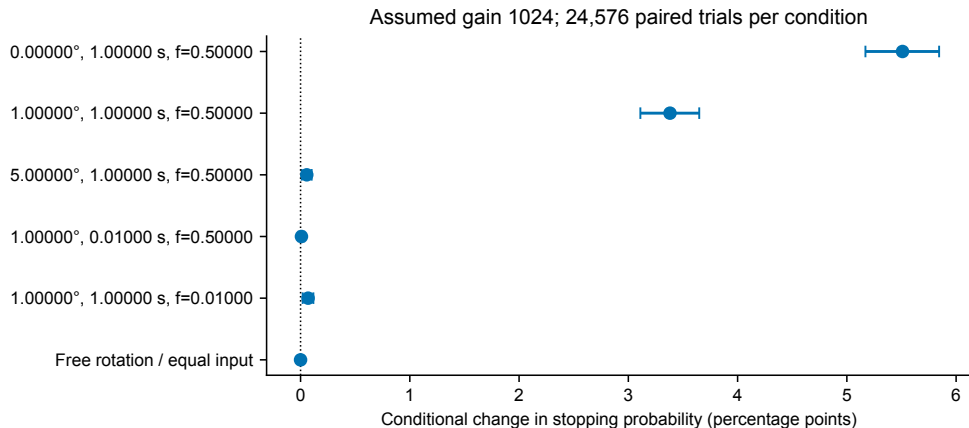


Fig. 4 Assumed downstream amplification preserves the ordering of selected molecular scenarios but does not calibrate a biological effect. Points show temporal-classifier paired stopping differences; bars are conservative intervals with at least 95% coverage under the simulation model. Each condition has 24576 pairs. All use eight selected connections, pool 10000, gain 1024 and chemical lifetime 200.00000 ms. Electric-field corrections are absent.

therefore describe the selected readout, with transition probabilities inherited from quantum dynamics within the cycle. This is consistent with the population-inheritance mechanism in Wong et al. [7], not evidence against it. Their Zeeman-basis projection, radical pair, photocycle and anisotropy observable differ from the present H-basis dark-cycle contrast. We have not reproduced their model in a matched limit and do not claim a general improvement in molecular realism. The rapid-renewal counterexample further rules out a universal statement that coherence is always irrelevant.

Nuclear population relaxation can dominate inter-cycle loss without being a single phenomenological T_1 . The Hamiltonians are anisotropic, several chemical states are involved, and the measured model observable is a reaction yield. A projection discarding off-diagonal terms at a selected boundary does not show that all relaxation is longitudinal in a single laboratory basis. The spectra, waiting maps and yield comparisons provide complementary diagnostics. Determining a biological longitudinal relaxation time would require specifying the prepared mode, observed nucleus, chemical state and motion ensemble.

The field comparisons distinguish mechanical alignment from changes in spin interactions. Strong orientational bias leaves enough motion to erase history in the free cases, while a deformable restraint changes a retained bound contrast. Electronic susceptibility is small in the selected frozen geometries. This ordering is not a universal ranking of membrane effects: protein geometry, protonation, environmental polarisation, field gradients, reaction kinetics and local noise spectra can respond differently. In particular, the small electronic-only yield corrections have not been shown to exceed uncertainty in those omitted mechanisms. The well calculation establishes a response of an assumed restraint, not a new native dissociation barrier or field-dependent binding lifetime.

The main model uncertainty is broader than numerical convergence. Only two nuclei are retained. Reaction rates and triplet preparation are imposed; the initial radical-pair identity and the omission of additional oxygen-electron relaxation constrain applicability. The short oxidised-state bath is transferred across states. Free and bound motion are coarse grained, the fast-wobble approximation becomes less secure at larger angles, and the local bath may overlap with the additional noise. Numerical validation addresses implementation and truncation, not these physical choices. In the binding sweep, selected $L = 2/3$ changes reach 3.06281×10^{-7} in Δq ; an independent finite-state noise check differs from the reduced H map by up to 0.59864% at 5.00000° . That finite-state comparison is neither an exact Ornstein–Uhlenbeck reference nor a full-cycle error bound. The electric combined-arm cutoff discrepancy is smaller, but does not remove physical-model uncertainty.

At the neural interface, evidence from light-stimulated fly neurons and redox-sensitive channels [11–13, 15, 16] supports candidate steps rather than the complete proposed pathway. The molecular yield has not been converted into a measured local chemical flux, channel-state change and release-probability change in the same preparation. A majority classifier in the synthetic circuit is an aggregation rule, not evidence that individual synapses make conscious stop/continue decisions. Classical accumulator models provide accounts of readiness-potential-like averages [27], and their interpretation remains debated [28]. Our calculation contains no electroencephalographic observation model, no empirical readiness-potential fit and no test of conscious veto or free will.

A discriminating experiment would independently vary angular restraint and binding dwell time while retaining the same chemical preparation and measuring a later product yield. The present model predicts loss on releasing the restraint under its slow-cycle conditions, even if a single-reaction magnetic response remains. Joint measurement of bound fraction, angular fluctuation spectrum and yield contrast would test the molecular mechanism before introducing a neural gain. A subsequent local chemical-to-channel calibration could then establish whether a retained contrast is large enough at a specified synapse or cell. Each proposed test follows from a model dependency and remains a prediction awaiting experiment.

5 Conclusion

Within a matched two-nucleus reaction model, free rotation eliminates the resolved slow-cycle nuclear-history contrast, while reversible binding permits retention that depends on dwell time and angular fluctuations. Electric orientation alone does not restore the free history; electronic response and deformation of an assumed restraint modify the retained bound contrast. The slow-cycle readout is reproduced by joint populations at the H boundary, with a rapid-renewal counterexample delimiting that approximation. The practical consequence is that neither a fixed-orientation lifetime nor a neural amplification example establishes a brain mechanism. Motion-resolved preparation and readout, followed by a measured local transduction, are the next tests required by this conditional model.

Supplementary information

Online Resource 1 contains derivations, parameter provenance, complete selected numerical tables and validation limits. Online Resource 2 contains compact frozen results, the reference operator archive and a bounded reproduction script. They accompany this local revision as separate files.

Declarations

Funding. This research received no external funding.

Competing interests. The author declares no competing interests.

Ethics approval and consent. This study reports computational modelling and analysis of existing scientific literature; no new human or animal experiments were conducted.

Author contributions. H.W. is the sole author and is responsible for the study and manuscript. Detailed contribution attribution and final approval of this revision are retained for author confirmation before submission.

Data and code availability. Computational source and the compiled manuscript and supplement are available at <https://github.com/deeptell-inc/cryptochrome-neural-history>. It excludes simulation results and large data. Online Resource 2 is supplied separately with this revision and contains the compact data and scripts needed to regenerate the reported figures, tables and selected numerical checks. It is excluded from the public source repository and has no public data deposit or persistent identifier. Complete upstream trajectories and electronic checkpoints are not included in the bounded archive; its README states the resulting reproduction limits.

Use of AI-assisted tools. Generative AI tools assisted with coding, computational checking and manuscript drafting and revision. Scientific Agent Skills supplied procedural writing and verification guidance [29]. The author reviews the generated outputs and assumes responsibility for the accuracy, integrity and final content of the work. Final human verification of this revision remains required before submission; AI-assisted checks do not constitute that approval.

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